

Chemical Equilibrium

* Relationship between ΔG° and Equilibrium Constant

$$\Delta G = \Delta G^\circ + RT \ln Q$$

At Equilibrium, $\Delta G = 0$,
 $Q = K$ (Equilibrium)

$$\Rightarrow 0 = \Delta G^\circ + RT \ln K$$

$$\Rightarrow \Delta G^\circ = -RT \ln K$$

$$\Delta G^\circ = -2.303 RT \log K$$

* Physical Equilibrium

① Solid-Liquid Equilibrium

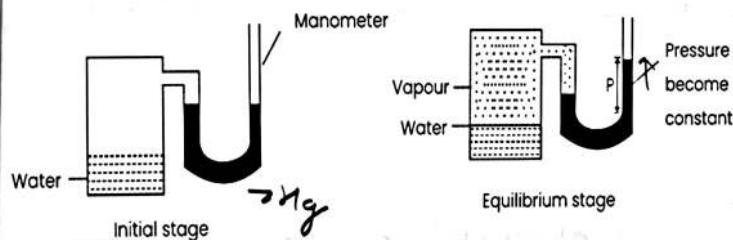
If a piece of ice is placed in a thermos & allowed to melt and weight of ice and liquid is measured. Now if we close the thermos flask, we observe that the weight of ice and liquid water remains constant with the passage of time.

i.e. - Rate of melting of ice = Rate of freezing of water

② Liquid Vapour Equilibrium

When liquid water is placed in a closed container, it starts evaporating increasing vapour pressure. This is shown by a rise in mercury level in a manometer. Eventually a constant pressure is reached, indicating equilibrium.

At the molecular level, evaporation increases vapour molecules, which collide with the liquid and condensation. Initially, evaporation is faster than condensation, but over time, both rates become equal. This balance maintains constant pressure - the state of liquid-vapour equilibrium.



③ Solid-Vapour Equilibrium

If we take Iodine (I_2) in a closed container. We notice that at the initial stage, Iodine sublimes in violet coloured vapours. With the passage of time, the violet colour becomes more intense i.e. sublimation takes place rapidly. But again intensity of violet colour becomes constant. At this stage
 Rate of Sublimation = Rate of deposition
 (I_2 solid) (I_2 vapour).

④ Solid-Solution Equilibrium

If we take a glass of water and add a solute (NaCl) in it, it dissolves in it. This solution is known as unsaturated solution. But if we keep adding the solute, it starts appearing in the solution. That means no more solute can be dissolved in the solution. That means no more solute can be dissolved in solution, at particular temperature. This is called saturated solution. At this - solute + solvent $\rightleftharpoons$ Solution
 • Rate of dissolution = Rate of crystallisation

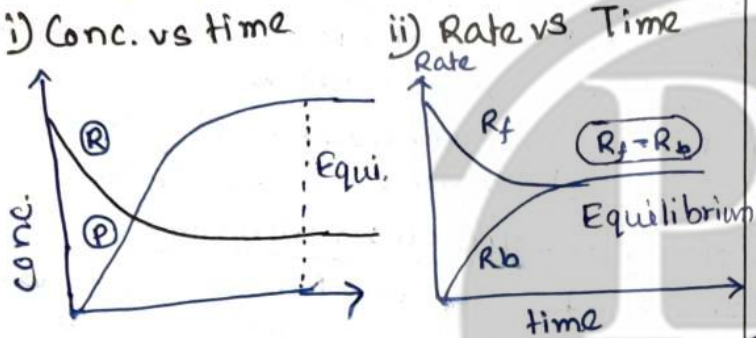
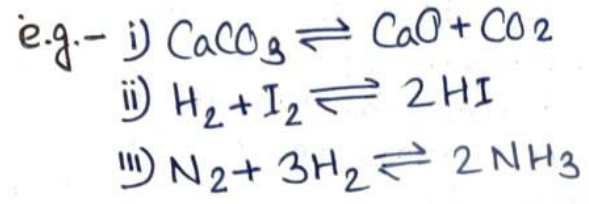
⑤ Gas solution Equilibrium

If we open a soda bottle the gas (CO_2) fizzes out from the bottle that means, the gas is dissolved in the bottle under pressure. There is equilibrium between Gas $\rightleftharpoons$ Liquid.

This is in accordance with Henry's Law. "States that mass of gas dissolved in a liquid is directly proportional to the pressure of the gas present above the liquid surface (m.p.)"

* Chemical Equilibrium

In case of reversible reactions, the stage at which concentration of reactants and conc. of product becomes constant is called chemical equilibrium. At this stage, Rate of forward reaction becomes equal to the rate of backward reaction.



* Dynamic nature of Chemical Equilibrium

As we know at equilibrium, concentration, temp, & pressure remains constant it seems the chemical reaction has passed or static but in reality process is continuously running. It is said to be dynamic in nature. For example in case of Haber's Process,



At equilibrium, the concentration of all the reactants and products remain constant at particular temperature. let us change the H_2 with isotope D_2 , then at equilibrium



Again the concentration remains same. Again let us mix H_2 &

D_2 in a reaction then we find at equilibrium different isotopic mixture. i.e. $\{\text{H}_2, \text{D}_2, \text{H-D}, \text{NH}_3, \text{ND}_3, \text{NHD}_2, \text{NH}_2\text{D}\}$. It proves that reaction is still running i.e. equilibrium is dynamic in nature.

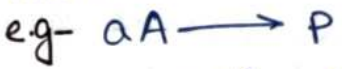
* Characteristics of Equilibrium

The important characteristics of equilibrium are as follows -

- i) In chemical equilibrium, at a given temperature, the measurable properties like pressure, concentration, density or colour remain constant.
- ii) Equilibrium can be attained from either side.
- iii) Equilibrium remains unaffected if the reaction occurs in presence of catalyst. However, equilibrium can be attained in a lesser time by the use of a catalyst.
- iv) Equilibrium is dynamic in nature (i.e. the reaction does not stop but continues in both the direction with same speed).
- v) The chemical equilibrium will continue indefinitely if the conditions such as temperature and concentration are not changed.

* Law of mass action

- This is given by Guldberg and Waage.
- It states that "rate of any chemical reaction is directly proportional to the active mass of the reactants raise to power of their stoichiometric co-efficient."



$$\text{Rate} \propto [\text{A}]^a \text{ or Rate} = k[\text{A}]^a$$

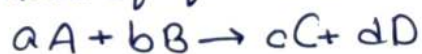
- Active mass means concentration.

Let us consider a chemical reaction at equilibrium.



Then according to law of mass action

Rate of forward reaction (R_f)



$$R_f = K_f [A]^a [B]^b \quad \text{--- (1)}$$

Rate of backward reaction (R_b)



$$R_b = K_b [C]^c [D]^d \quad \text{--- (2)}$$

• At equilibrium, $R_f = R_b$

$$\Rightarrow K_f [A]^a [B]^b = K_b [C]^c [D]^d$$

$$\Rightarrow \frac{K_f}{K_b} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$$\Rightarrow K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$$\frac{K_f}{K_b} = K_c$$

$K_c = \text{Equilibrium constant}$

Thus, we can say that, ratio of multiplication of

concentration of products raised to power their stoichiometric coefficient to the multiplication of concentration of reactants raised to power their stoichiometric coefficient is constant. This is law of chemical equilibrium.

This is known as Equilibrium constant.

* Characteristics of equilibrium constant

1) Equilibrium constant does not depend on initial concentration of reactants. It only depends upon temperature.

2) If we divide chemical equation by 'n' which is at

equilibrium, then the rate constant $K = K^{1/n}$

③ If we multiply the equilibrium equation with 'n' then, new equilibrium constant $K = K^n$

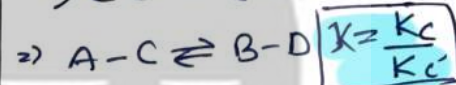
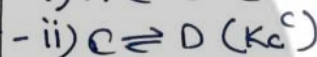
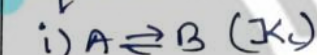
④ If we reverse the equilibrium equation, then the relation between old K_c & new K_c is
i) $A \rightleftharpoons B$ K_c
ii) $B \rightleftharpoons A$ $1/K_c$

$$\text{i.e. } K_c = \frac{1}{K_c}$$

⑤ If we add two equilibrium equations, then
i) $A \rightleftharpoons B$ (K_c)
+ ii) $C \rightleftharpoons D$ (K_c')



⑥ If we subtract two equations, equilibrium equation subtract, then



* Equilibrium constant for gas phase reaction (K_p)

→ Suppose a chemical reaction



then according to law of chemical equilibrium

$$K_p = \frac{P_C^c P_D^d}{P_A^a P_B^b} \quad \text{here, } P = \text{partial pressure of the gas}$$

* Relation between K_p & K_c

Consider a reaction $aA + bB \rightleftharpoons cC + dD$

$$K_p \text{ can be written as } K_p = \frac{P_C^c P_D^d}{P_A^a P_B^b}$$

We know that, according to ideal gas equation, $PV = nRT$

$$\Rightarrow P = \frac{n}{V} RT \Rightarrow P = CRT$$

$$P_C^c = (C_C RT)^c \quad P_D^d = (C_D RT)^d$$

$$P_A^a = (C_A RT)^a \quad P_B^b = (C_B RT)^b$$

* by replacing the value of P in equation, we get.

$$K_P = \frac{(C_C RT)^c (C_D RT)^d}{(C_A RT)^a (C_B RT)^b}$$

$$K_P = \frac{C_C^c (RT)^c C_D^d (RT)^d}{C_A^a (RT)^a C_B^b (RT)^b}$$

$$K_P = \frac{C_C^c C_D^d (RT)^c (RT)^d}{C_A^a C_B^b (RT)^a (RT)^b}$$

$$K_P = \frac{C_C^c C_D^d (RT)^{c+d}}{C_A^a C_B^b (RT)^{a+b}}$$

$$K_P = \frac{C_C^c C_D^d}{C_A^a C_B^b} RT^{(c+d)-(a+b)}$$

$$K_P = \frac{C_C^c C_D^d}{C_A^a C_B^b} (RT)^{\Delta n_g} \quad \left[\begin{array}{l} \text{change in no. of} \\ \text{moles,} \\ \Delta n_g = (c+d) - (a+b) \end{array} \right]$$

$$= \frac{[C]^c [D]^d}{[A]^a [B]^b} (RT)^{\Delta n_g}$$

$$K_P = K_C (RT)^{\Delta n_g}$$

* Unit of Equilibrium Constant

* If no of moles of reactant and no of moles of product are not equal then the unit can be -

K_{eq} has no unit.

* If not of moles of reactant and no of moles of product are not equal then the unit can be derived from following formula.

$$\text{i.e.} - K_C = (\text{mol} \cdot \text{L}^{-1})^{\Delta n_g}, \quad K_P = (\text{atm})^{\Delta n_g}$$

* Application of Equilibrium Constant

① To determine the extent of Reaction

* If the value of K_{eq} is more than 10^3 then the amount of product is more in reaction i.e. reaction is about to end.

* If the value of K_{eq} is less than 10^{-3} then the amount of reactant is more i.e. in initial stage of the reaction.

* If the value of K_{eq} is between 10^3 to 10^{-3} , then there is considerable amount of reactant & product.

② To determine the Direction of Reaction

* To determine the direction of the reaction we introduce a new term, called as reaction quotient (Q).

* It is also the ratio of molar conc or partial pressure of reactant at any time. i.e. → For a reaction



$$Q = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

* i) If $Q = K_{eq}$, then reaction is at equilibrium.

ii) If $Q > K_{eq}$, then reaction proceeds in backward direction

iii) If $Q < K_{eq}$, then reaction proceeds in forward direction.

* Le Chatelier's Principle

* This principle was given by Louis Le Chatelier, Russian Chemist.

* It states that "if a system in a equilibrium is subjected to change pressure, concentration, temperature etc. then equilibrium shifts in a direction which tends to reduce the effect of change."

① Effect of Concentration Change

Consider an example, $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$
If we added any reactant (H_2 , I_2) in a mixture, that is concentration of reactant is increased, then according to Le Chatelier's principle, equilibrium will shift in a direction where conc. of reactant is decreasing i.e. in forward direction & vice versa.

• We can say that,

→ If the concentration of reactant is increased, reaction shifts in forward direction.

→ If the concentration of product is increased, reaction shifts in backward direction.

② Effect of Pressure

Consider a reaction, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$
[For no. of moles of gas]

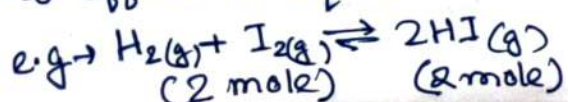
If we are increasing the pressure of the system, i.e. we are increasing the no. of moles of the system, then according to Le Chatelier's principle, equilibrium will shift in a direction where no. of moles of gas are decreasing & vice versa.

• We can say that.

→ At high pressure, equilibrium favours less no. of moles.

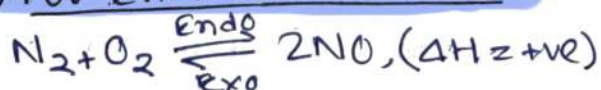
→ At low pressure, equilibrium favours more no. of moles.

• If $\Delta n_g = 0$, the pressure has no effect on equilibrium.



③ Effect of Temperature

a) For Endothermic Reaction



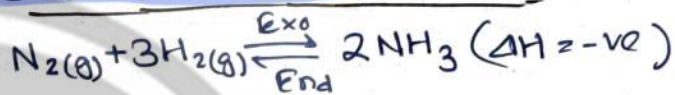
If the temperature is increased, then the equilibrium will shift in forward direction (Endothermic Rxn).

If 'T' is decreased, then equilibrium will shift back in backward direction (Exothermic Rxn).

i.e. → High Temperature favours Endo.

Low Temperature favours Exo rxn.

b) For Exothermic Reaction



→ If temperature is increased, Equilibrium shifts in backward direction (Endothermic Rxn).

→ If temperature is decreased, Equilibrium shifts in forward direction.

④ Effect of adding Inert gas.

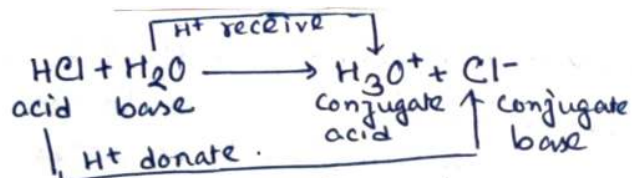
a) At Constant Volume

We add inert gas to the reaction mixture, total pressure is increased, but it does not affect on conc, temp, pressure of reaction, then there is no effect on equilibrium.

b) At Constant Pressure

At constant pressure, if we add inert gas to the rxn mixture, total volume increases. No. of moles of gas in per unit volume decreases.

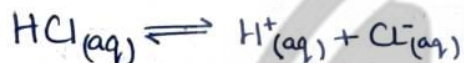
→ If no. of moles of gas decrease, then equilibrium will shift in that direction where no. of mole increases.



- Thus, same compound can behave acid and base in different reaction.
- If Acid is strong then its conjugate base will be weak & if base is strong then its conjugate acid will be weak
- Conjugate acid base pair differs only with one H^+ .

* Ionization of Acid and Bases

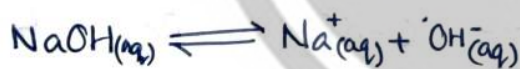
* Acid dissociate in water to produce H^+ ions. e.g. $\rightarrow \text{HCl}$



$$K_a \text{ or } K_c = \frac{[\text{H}^+][\text{Cl}^-]}{[\text{HCl}]}$$

K_a is ionisation constant of acid or acid dissociation.

* Base dissociate to produce OH^- in water. e.g. $-\text{NaOH}$



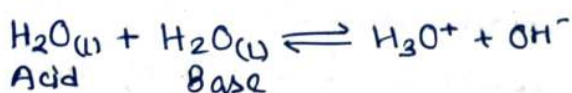
$$K_b = \frac{[\text{Na}^+][\text{OH}^-]}{[\text{NaOH}]}$$

K_b is the ionization constant of base or Base dissociation constant.

- Strong acid: $\text{HCl}, \text{HI}, \text{H}_2\text{SO}_4, \text{HNO}_3$
- Weak acid: $\text{CH}_3\text{COOH}, \text{H}_2\text{CO}_3, \text{H}_3\text{BO}_3$ etc
- Strong Base: $\text{NaOH}, \text{KOH}, \text{Mg}(\text{OH})_2, \text{LiOH}$
- Weak base: $\text{NH}_4\text{OH}, \text{Al}(\text{OH})_3$ etc.

* Ionic Product of Water

If we dissolve water in water, the one water molecule behave as acid and other molecule behave as base.



∴ Equilibrium Constant, $K_c = \frac{[\text{H}_3\text{O}^+][\text{OH}^-]}{[\text{H}_2\text{O}]^2}$

As we know that concentration of pure liquid is constant (Unity)

$$\text{i.e.} \rightarrow K_c \times [\text{H}_2\text{O}]^2 = [\text{H}_3\text{O}^+][\text{OH}^-]$$

K_w is constant name water dissociation constant. or Ionic product of water. Experimentally it has been found that at 298K. conc. of $[\text{H}_3\text{O}^+] = [\text{OH}^-] = 1 \times 10^{-7} \text{M}$

$$\text{i.e.} \rightarrow K_w = 10^{-7} \text{M} \times 10^{-7} \text{M}$$

M = Molar
= mol l^{-1}

$$K_w = 10^{-14} \text{M}^2$$

* Constant of pH

The term pH is derived from a word "potenz" means power of hydrogen. This concept was given 'Sorenson' to express the concentration of $[\text{H}^+]$ or $[\text{H}_3\text{O}^+]$ in any solution. It is defined as "Negative logarithm of concentration of $[\text{H}^+]$ or $[\text{H}_3\text{O}^+]$ in any solution."

• Mathematically,

$$\text{pH} = -\log [\text{H}^+] \text{ or } \frac{1}{\log [\text{H}^+]}$$

We know that, at 298K, $[\text{H}^+] \text{ or } [\text{H}_3\text{O}^+] = 1 \times 10^{-7} \text{M}$.

$$\text{pH} = -\log 10^{-7} = (-7)(-1 \log 10) = 7 \log 10 = 7$$

It is considered a neutral solution.

• In the same way, if

i) $\text{pH} < 7$, it is acidic solution

ii) $\text{pH} > 7$, it is basic solution.

iii) $\text{pH} = 7$, it is neutral solution.

* Log Value

$$\log 1 = 0, \log 2 = 0.30, \log 3 = 0.47$$

$$\log 4 = 0.60, \log 5 = 0.69, \log 6 = 0.77$$

$$\log 7 = 0.84, \log 8 = 0.90, \log 9 = 0.95$$

$$\log 10 = 1$$

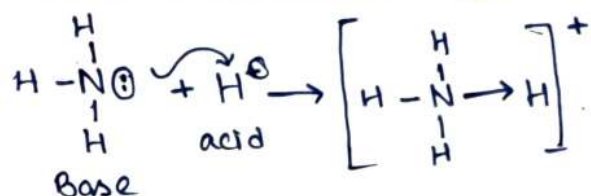
* Concept of Acid & Base

* Lewis Acid and Bases

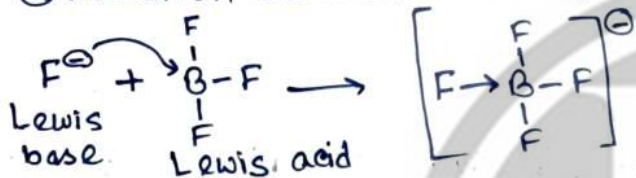
According to this concept, Acid is a substance that accepts a lone pair of e^- . Base is a substance that donates a lone pair of e^- .

• For example:-

① Reaction between NH_3 & H^+



② Reaction between F^- & BF_3



* All Lewis bases behave as

Brownstred base.

* sp^2 hybridised or cation $\rightarrow$ Lewis Acid

* lone pair or anion $\rightarrow$ Lewis base.

* Relationship between pH & pOH

Just like pH, we can define pOH also. i.e. $pOH = -\log[OH^-]$

Applying log on b/s on Ionic product at 298K, $K_w = [H^+][OH^-]$,

$$\begin{aligned} \log K_w &= \log([H^+][OH^-]) \\ &= \log[H^+] + \log[OH^-] \end{aligned}$$

* We know that,

$$\begin{array}{l} pH = -\log[H^+] \quad pOH = -\log[OH^-] \\ \Rightarrow \log[H^+] = -pH \quad \Rightarrow \log[OH^-] = -pOH \end{array} \quad \log K_w = -pK_w$$

* then, $-pK_w = -pH - pOH$

$$\Rightarrow pK_w = pH + pOH \quad \text{--- (1)}$$

$$pH + pOH = 14$$

At, 298K
 $pK_w = 14$

* Degree of dissociation of weak Acid

As we know weak acid do not dissociate completely in water.
e.g. $\rightarrow CH_3COOH$ (Acetic acid)



* At time '0' concentration of weak acid 'C'
At time 't' suppose $C\alpha$ get dissociated from 'C',

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]} = \frac{[C\alpha][C\alpha]}{C - C\alpha} = \frac{C\alpha^2}{C(1-\alpha)}$$

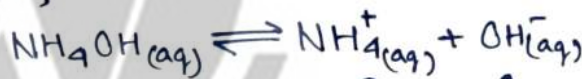
$$K_a = \frac{C\alpha^2}{1-\alpha}$$

* For weak acid $1 \gg \alpha$, α can be neglected in denominator.

$$K_a = C\alpha^2 \Rightarrow \alpha^2 = \frac{K_a}{C} \Rightarrow \alpha = \sqrt{\frac{K_a}{C}}$$

* Degree of dissociation of weak base

Similarly, for a weak base like NH_4OH ,



$$K_b = \frac{(C\alpha)(C\alpha)}{C - C\alpha} = \frac{C\alpha^2}{C - C\alpha} = \frac{C\alpha^2}{1 - \alpha}$$

* For weak base, $1 \gg \alpha$

$$\text{then } K_b = C\alpha^2 \Rightarrow \alpha^2 = \frac{K_b}{C} \Rightarrow \alpha = \sqrt{\frac{K_b}{C}}$$

* Salt Hydrolysis

Salt hydrolysis is defined as the process in which a salt reacts with water to give back the acid and base



Thus hydrolysis may be considered as reverse of neutralization. Thus depending upon the relative strength of the acid

and base produced, the salt are divided into four categories.

① Salt of strong acids & strong bases

e.g. $\rightarrow \text{NaCl}, \text{NaNO}_3, \text{Na}_2\text{SO}_4, \text{KCl}, \text{KNO}_3, \text{K}_2\text{SO}_4$.

As an illustration, let us discuss the hydrolysis of NaCl we may write



② Salt of weak acids & strong bases

e.g. $\rightarrow \text{CH}_3\text{COONa}, \text{Na}_2\text{CO}_3, \text{K}_2\text{CO}_3, \text{Na}_3\text{PO}_4$ etc.

* Hydrolysis of sodium acetate (CH_3COONa) $\rightarrow$



③ Salt of strong acid & weak bases

e.g. $\rightarrow \text{NH}_4\text{Cl}, \text{CuSO}_4, \text{NH}_4\text{NO}_3, \text{AlCl}_3, \text{CaCl}_2$ etc.

* Hydrolysis of NH_4Cl



④ Salt of weak acid & weak bases

e.g. $\rightarrow \text{CH}_3\text{COONH}_4, (\text{NH}_4)_2\text{CO}_3, \text{AlPO}_4$

* Hydrolysis of $\text{CH}_3\text{COONH}_4$



* Application of Buffer Solution

i) Normal blood has pH of 7.4. This pH is maintained by buffer action following two buffers.

$\rightarrow$ Buffer of H_2CO_3 & NaHCO_3

ii) Buffers find numerous application in industrial process like manufacture of paper, drugs, dyes, inks etc.

iii) Buffer solutions are applied

for digestion process of food.

iv) These are used as preservative of foods and fruits.

v) Buffers can be used for preservation of many dairy and agricultural products.

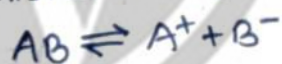
* Buffer Capacity (Φ)

Buffer capacity is quantitatively defined as the number of moles of acid or base added in 1L of solution to change the pH by unity

$$\Phi = \frac{\text{Number of moles of acid or base added to 1L of buffer}}{\text{Change in pH}}$$

* Common Ion Effect

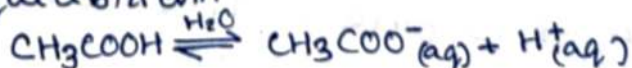
If in a ionic equilibrium of any substances we add salt having common ions, then the equilibrium will shift in backward direction, this is called common ion effect. Consider a following equilibrium



If we add any salt having (A^+) or (B^-) then concentration of common ion will increased in products, therefore according to Le Chatelier principle, equilibrium will shift in backward direction.

* For Example

i) In case of ionic equilibrium of CH_3COOH , when CH_3COOH is placed in water then there will be equilibrium

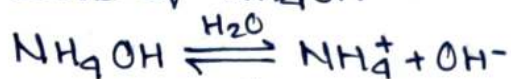


If we add salt $\text{CH}_3\text{COO}^- \text{Na}^+$, then as we know CH_3COONa will dissociate completely.



Therefore, CH_3COO^- is a common ion which will increase the conc. of CH_3COO^- in an equilibrium, decomposition of CH_3COOH is further decreased. because equilibrium will shift in backward direction.

ii) Consider the ionic equilibrium in case of NH_4OH



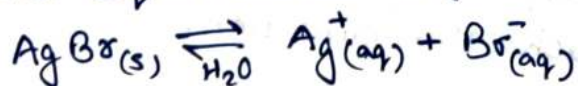
If we add salt NH_4Cl to this solution then NH_4Cl will dissociate completely.



Therefore conc. of NH_4^+ will increase in product which will shift in backward direction, according to Le-Chatelier principle.

* Solubility Product (K_{sp})

Some salt like, AgBr , AgCl , PbCl_2 etc. dissolve to very small extent in water are called sparingly soluble salt. When the salts are dissolved in water there exist equilibrium. e.g.



According to Law of chemical Equilibrium, $K = \frac{[\text{Ag}^+][\text{Br}^-]}{[\text{AgBr}]}$

AgBr is pure solid, hence its conc. will remain const.

$$K \times [\text{AgBr}] = K_{sp}$$

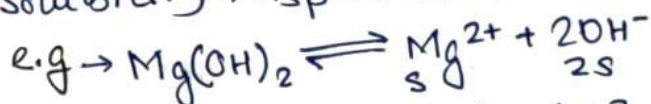
$$\Rightarrow K_{sp} = [\text{Ag}^+][\text{Br}^-]$$

K_{sp} is solubility product.

It is defined as product in case of sparingly soluble salt raised to power their stoichiometry.

$$K_{sp} = [\text{Ag}^+][\text{Br}^-]$$

In case of sparingly soluble salt concentration is taken as molar solubility $\rightarrow K_{sp} = s \times s = s^2$



$$K_{sp} = [\text{Mg}^{2+}][\text{OH}^-]^2 = s(2s)^2 = 4s^3$$



$$K_{sp} = [\text{Ca}^{2+}]^3 [\text{PO}_4^{3-}]^2 = (3s)^3 (2s)^2 = 27 \times 4s^5$$

* Buffer Solution

"Buffer solution is that type of solution whose pH does not change on addition of small amount of acid and base"

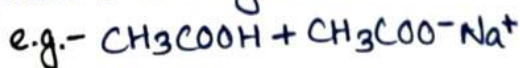
This property of a solution of resulting pH change is called buffer action.

* Type of Buffer Solution

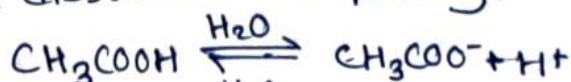
i) Acid buffer ii) Basic buffer.

i) Acid Buffer

It is the mixture of weak acid & its conjugate base (Salt of weak acid & strong base).



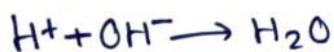
As we know, CH_3COOH is dissociated completely and CH_3COONa will dissociate completely.



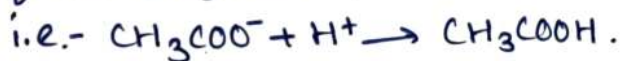
Due to common ion effect dissociation of CH_3COOH is further

decreased.

On addition of base
i.e. OH^- , it combines with H^+ to
form H_2O



On addition of acid i.e. H^+ ,
this H^+ combines with CH_3COO^- to
form CH_3COOH .

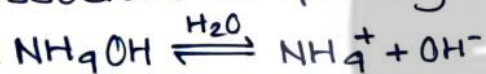


Hence water is neutral, then
pH will not change.

ii) Basic Buffer

Basic buffer is a mixture of weak
base and its conjugate acid (salt
of weak base and strong acid).
e.g. $\text{NH}_4\text{OH} + \text{NaCl}$.

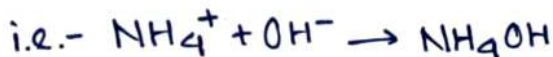
NH_4OH dissociates in completely
in a water and salt NH_4Cl
dissociate completely



On addition of acid i.e. its
combines with OH^- to form H_2O
i.e. $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$

Water is neutral, thus pH will
not change.

On addition of base i.e. OH^- , it
combines with NH_4^+ to form weak
base NH_4OH .



Therefore, pH will not change.